

Daniel Coolen

Senior Data Engineer | Data Platforms & Data Reliability

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SUMMARY

Senior Data Engineer with 10+ years of hands-on experience designing and operating **AWS-native data platforms**. Proven track record in building **reliable, scalable data infrastructure** using **Python** and **SQL**, specializing in **Apache Spark**, **Kafka**, **Redshift**, and **Airflow**. Focused on deterministic processing, **data contracts**, and **replay-safe transformations** for regulated and high-concurrency environments.

I have a deep background in **cloud-based architecture**, **ML infrastructure**, and **data reliability engineering**.

I prioritize **data integrity**, **compliance**, and **scalability**, and have successfully led platform-level decisions that reduce operational risk and improve system performance. Known for **designing systems** that ensure long-term sustainability, **data lineage**, and **auditable pipelines**.

SKILL

Programming & Data Engineering

Python (PySpark, production pipeline development, performance optimization, testing with PyTest)

SQL (advanced analytics, query optimization, window functions, warehouse tuning)

Apache Spark (batch processing, structured streaming)

Apache Kafka (event-driven pipeline design, real-time streaming)

Apache Airflow (DAG orchestration, workflow management, failure recovery)

Cloud & Infrastructure (AWS-First)

AWS (S3, Redshift, EMR, Glue, Lambda, IAM, CloudWatch)

Terraform (Infrastructure as Code)

Docker (containerization for environment consistency)

CI/CD (GitHub Actions, GitLab CI for data pipelines)

AWS Cost Optimization (cost management for cloud data workloads)

Monitoring & Logging (CloudWatch, Datadog)

Data Warehousing & Modeling

Amazon Redshift (performance tuning, query optimization)

dbt (modular transformations, testing, schema documentation)

Dimensional Modeling (star/snowflake schema, data mart design)

Data Contracts (schema versioning and integrity enforcement)

Data Quality & Reliability

Great Expectations (data quality validation frameworks)

dbt tests, PyTest (automated testing for data pipelines)

Data SLAs, observability frameworks, and pipeline monitoring

Data Lineage (traceability, reproducibility, audit defense)

Governance-aware workflows (HIPAA compliance)

ML & AI Data Infrastructure

ML pipeline orchestration (Airflow for ML features)

Feature engineering pipelines (data preparation for ML models)

MLflow (model tracking, artifact management)

LLM data ingestion workflows (feature extraction for ML models)

Vector/embedding data processing for training AI models

Business Intelligence

Tableau, Looker (data visualization, dashboarding, metrics validation)

Stakeholder-facing dashboards for operational and executive decision-making

EXPERIENCE

Surescripts | Chicago, IL (Remote)

01/2022 – Present

Senior Data Engineer

- Own critical components of a national **healthcare data exchange platform**, building **AWS-native data infrastructure** where prescription transaction accuracy directly impacts **regulatory reporting** and **pharmacy network operations**.
- **Re-architected ingestion pipelines** using **Apache Spark (3.x)** and **Amazon S3**, introducing **deterministic transformations** and **idempotent processing** to ensure consistent, auditable data across regulated environments.
- Designed **distributed processing frameworks** handling **10TB+ daily transactions** with **Python** and **SQL**, ensuring **audit replayability** and supporting **data integrity** through **version-controlled schema evolution**.
- Embedded **HIPAA-aligned encryption** with **S3 server-side encryption** and **IAM least-privilege access policies**, securing **healthcare data** at rest and in transit across multiple ingestion and transformation pipelines.
- Introduced **contract-level schema validation** and **referential integrity enforcement** during **ingestion**, reducing reporting defects by **~40%** and preventing silent data corruption downstream.
- **Optimized Apache Airflow orchestration** by isolating failure domains and implementing **SLA-driven alerting** to minimize **data freshness regressions** during peak **pharmacy transaction windows**.
- **Re-optimized Redshift cluster performance**, using **SQL** to fine-tune **distribution keys**, **sort keys**, and **WLM queues** to segregate compliance-critical workloads from exploratory analytics, improving **peak-hour stability** by **~45%**.
- Introduced **CI/CD regression validation gates**, embedding **PyTest** and **dbt tests** to **block schema-breaking deployments** before reaching production, ensuring **compliance** and data integrity at scale.
- Led **architectural decisions** prioritizing **data lineage traceability** over minor latency optimizations, promoting **compliance-first tradeoffs** in platform evolution discussions.
- Formalized **reproducibility standards** across transformation layers, enforcing **deterministic logic** and **controlled schema migrations** to support **audit defense** and **incident replayability**.
- Designed **ML feature pipelines** using **Python** and **SQL**, ensuring **training** and **inference consistency** under evolving schemas and incremental ingestion updates.
- **Collaborated with Compliance leadership** during **external audit preparations**, delivering detailed **transformation trace artifacts** and **lineage documentation** to meet strict regulatory deadlines.
- Built **reusable Spark transformation frameworks** and **standardized data contract enforcement** patterns, adopted across multiple domain teams to improve **scalability** and **cross-team collaboration**.
- Elevated **engineering standards** by embedding **data reliability** practices such as **validation**, **lineage**, and **traceability** into the core design principles of all data systems.
- **Diagnosed production incidents** by tracing failures through **distributed Spark jobs** and **SQL queries**, reducing **MTTR** by **~35%** through **root-cause analysis** and structured **pipeline improvements**.

- Modeled **infrastructure scaling** based on transaction volume projections, influencing **compute allocation strategy** and **Redshift cluster resizing** to proactively accommodate growth.
- Introduced **performance benchmarking** for **batch vs. streaming architectures**, ensuring balanced tradeoffs between **real-time analytics** and **system reliability** while maintaining scalability.
- Created **auditable data pipelines** that ensure full **reproducibility** and **deterministic processing** across all data transformation layers to guarantee **consistent reporting** across all healthcare data sources.
- Built **cross-team adoption** of standardized **data contracts** and **validation frameworks**, integrating **Great Expectations** and **dbt** to enhance overall **data governance** and **quality** across teams.
- Enhanced the **engineering culture** by emphasizing **defensive data reliability engineering** practices and instilling a mindset of **compliance-first architecture** and operational accountability within the data engineering team.

Amplitude | San Francisco, CA

09/2018 – 12/2021

Data Engineer

- Spearheaded **data infrastructure** for a **multi-tenant analytics platform**, leveraging **Apache Kafka** and **AWS S3** to enable real-time event ingestion for thousands of **SaaS customers**, optimizing for high-velocity product changes.
- Enhanced **real-time analytics** by tuning **Kafka streaming pipelines** and **Spark transformations**, achieving sub-5-minute latency for **A/B testing** and **experiment-driven decision-making** across a **high-concurrency environment**.
- Led platform evolution from **batch-heavy analytics** to a **hybrid streaming + batch architecture**, using **Apache Airflow** for efficient DAG orchestration, ensuring high availability and low-latency data flows critical for product iteration.
- Reworked core **Spark** transformation layers (**PySpark**) to **isolate tenant workloads** and enforce backward-compatible **schema evolution**, reducing the need for costly historical rewrites and improving system scalability.
- Optimized **SQL-based aggregation models** in **Amazon Redshift**, adjusting **distribution keys** and **sort strategies** to enhance performance for high-concurrency **enterprise dashboards** during peak usage.
- Built and deployed **data quality checks** within **Spark jobs** using **Python** and **SQL**, stopping corrupted event payloads from affecting downstream aggregation and reporting, reducing **metric drift**.
- Designed **automated testing frameworks** in **dbt** and **PyTest**, ensuring **data contracts** and **schema integrity** through continuous validation and preventing schema-breaking changes from reaching production.
- Introduced **tenant-aware resource controls** within **Spark transformations**, reducing **cross-customer contention** during bursty analytics workloads and enabling a stable environment for diverse client demands.
- Contributed to **capacity modeling** for **Kafka brokers**, **Spark compute**, and **Redshift clusters**, aligning infrastructure growth with **sales projections** rather than reactive scaling, ensuring smooth customer onboarding.
- Led root-cause analysis for **freshness regressions** by identifying failures in **Kafka ingestion pipelines** and **upstream instrumentation** rollouts, improving **MTTR** by ~35% through systemic fixes.
- Enforced **data governance** policies with **event contract enforcement**, ensuring consistent **metric definitions** across the product, which improved product alignment between **engineering** and **data science** teams.
- Pioneered a **modular approach to ML feature pipelines**, leveraging **Apache Spark** and **AWS S3** to ensure **data consistency** across **training** and **inference pipelines**, reducing model training downtime by ~40%.
- Collaborated with **Product Managers** and **Data Science teams** to define **data pipelines** that directly supported product experimentation workflows, emphasizing **data reliability** and **quality assurance** at every stage.
- Established **best practices** for **cross-team collaboration**, adopting **standardized frameworks** for data contract enforcement, **data lineage** tracking, and **versioned schemas**, which led to improved **data governance**.
- Played a key role in **engineering culture** by championing **defensive data engineering** principles, promoting **scalable solutions**, **traceable data workflows**, and an emphasis on **long-term infrastructure reliability**.

Data Analyst

- Engineered large-scale **Hadoop** batch pipelines to process **2–3TB/day** of industrial **IoT telemetry** using **HiveQL** and **SQL**, supporting predictive maintenance analytics across 15+ enterprise deployments.
- Developed **Python**-based data validation frameworks detecting **sensor drift**, **null bursts**, and **timestamp anomalies**, ensuring high-quality telemetry data for downstream analytics.
- Strengthened **Hadoop ingestion** pipelines by embedding **reconciliation checks** and **validation logic** in **Hive**, increasing success rates from ~92% to 98% in a highly dynamic, sensor-driven environment.
- Designed **dimensional data models** (fact/dimension schemas) using **SQL** to standardize KPI definitions across clients, resolving discrepancies in telemetry data sources and unifying multi-client reporting standards.
- Collaborated with **Data Science** teams to streamline feature engineering workflows, implementing **schema alignment** and **dataset versioning** in **Python** and **Spark 2.x** to reduce model rework cycles.
- Contributed to early adoption of **Apache Spark 2.x**, migrating heavy aggregation workloads from **Hive** to **Spark**, reducing telemetry processing time by ~35% and improving cluster efficiency.
- Implemented controlled **schema evolution** practices within **Hive tables**, ensuring backward compatibility and preventing breaking changes in **ETL** pipelines.
- Optimized **distributed HiveQL queries**, fine-tuning **partitioning** and **indexing** strategies to reduce query latency by ~30% and improve the overall performance of shared **Hadoop clusters**.
- Supported the migration of **archive datasets** to **Amazon S3**, improving cost efficiency and providing scalable, hybrid storage access for downstream **analytics**.
- Worked with cross-functional teams to design and implement **automated testing frameworks** using **PyTest** and **dbt tests** to ensure consistent, reliable data pipelines.
- Designed and implemented **telemetry aggregation layers** using **SQL** and **Spark**, enabling time-series feature extraction critical for **predictive maintenance** scoring.
- Participated in architectural discussions on **distributed compute scaling**, favoring **incremental adoption of Spark** over full re-platforming to ensure system reliability in a high-growth startup environment.
- Modeled **data pipeline scaling** based on transaction growth projections, helping define **compute allocation** and **Hadoop cluster resizing** strategies before infrastructure limits were reached.
- Standardized repeatable batch workflows using **Oozie** and **Hive**, balancing flexibility for experimental **data science** work with **production-grade reporting** reliability.
- Led **root-cause analysis** for performance regressions, identifying failures in upstream **ETL pipelines** and preventing premature algorithmic adjustments by tracing issues through the **data processing pipeline**.
- Fostered a **data reliability culture** by promoting **data contracts**, **versioning**, and **cross-team collaboration**, enhancing transparency and ensuring data consistency across departments.

EDUCATION

Bachelor of Science in Computer Science

08/2011 – 05/2015

University of Illinois Chicago (UIC) – Chicago, IL